让蚂蚁变僵尸的真菌

The Fungus That Turns Ants into Zombies



蚂蚁为何会咬住叶片不松口?

英文原文

Deep in a Thai **rainforest**, a **carpenter** ant breaks from its **colony**'s **column** in the **lofty canopy**. **Swaying** on **unsteady** legs, it **wanders downward**, **as if intoxicated**, into **leafy undergrowth**. Around **midday** it **clenches** its **jaws** on a major **vein** underneath a leaf, about 25 centimetres above the ground.

By **dusk** the insect is **lifeless**, yet the bite holds **rigidly**. A fungus has destroyed its jaw **muscles**, so they cannot **loosen**. Two or three days later a **fungal** stalk pushes from its head and releases **infectious** spores on to the ground below, where foraging ants **tread**.

The **invader**, Ophiocordyceps unilateralis, is **fussy** about **location**. David Hughes and colleagues found the **bites** mostly on the **north-west** side of plants, where the air is **humid** and the temperature suits the fungus. When researchers moved infected ants higher, the fungus grew **abnormally**. Placed on the ground, the bodies **disappeared**.

In 2017 a Penn State team examined infected ants with **three-dimensional microscope** imaging. Fungal cells had formed an **intertwined lattice** around muscle fibres in the legs and jaw, a picture that **evokes** the strings of a **puppet**. Yet the fungus had not **infiltrated** the brain.

A Planet Earth episode, with David Attenborough's **narration**, showed one such ant. It helped **inspire** The Last of Us, a 2013 video game whose **makers envisioned** the parasite jumping to humans. The **hypothetical scenario** gave **gamers** a **fictional** nightmare.

The real parasite's roots run far deeper, into the **geological** record. A 48-million-year-old **fossil** leaf from Messel, in Germany, bears the same **characteristic** bite **scars**.

中文译文

在泰国雨林深处，一只木匠蚁脱离了它的蚁群队列，在高耸的树冠上（前行）。摇摇晃晃，腿脚不稳，它向下游荡，好像喝醉了一样，进入枝叶茂密的林下植被。正午前后，它将双颚紧咬在一片叶子下方的一条主叶脉上，离地面约25厘米。

到黄昏时，这只昆虫已无生命，那一口却仍牢牢地咬着。一种真菌摧毁了它的颚部肌肉，因此它们无法松开。两三天后，一根真菌菌柄从它头部顶出，将具有传染性的孢子释放到下方的地面上，那里有觅食的蚂蚁行走。

这个入侵者，也就是偏侧蛇虫草菌，对位置很挑剔。戴维·休斯和同事发现，这些咬合点大多在植物的西北侧，那里空气潮湿，且温度适宜真菌。研究人员把受感染的蚂蚁移到更高处时，真菌长得很反常。放在地面上，尸体则不见了。

2017年，宾夕法尼亚州立大学的一个团队检查了受感染的蚂蚁，用的是三维显微镜成像。真菌细胞形成了相互交织的网格，围在腿部和颚部的肌肉纤维周围，一幅使人想起木偶的提线的画面。不过，真菌尚未渗入大脑。

《地球脉动》的一集，伴随着戴维·阿滕伯勒的解说，就展示了这样一只蚂蚁。它帮助启发了2013年的电子游戏《最后生还者》，游戏的制作者们设想这种寄生菌跳到了人类身上。这个假想的情景给玩家带来了一场虚构的噩梦。

现实中的寄生菌，其根源要深得多，一直延伸到地质记录之中。一片4800万年前的叶子化石，出土于德国梅塞尔，上面带有同样典型的咬合疤痕。

词汇与表达

rainforest n.
雨林

carpenter n.
木匠

colony n.
群
its colony's column

column n.
队列

lofty adj.
高耸的

canopy n.
树冠
in the lofty canopy

Swaying v.
摇摇晃晃

unsteady adj.
不稳
on unsteady legs

wanders v.
游荡
it wanders downward

downward adv.
向下

as if 连词短语
好像

intoxicated adj.
喝醉了
as if intoxicated

leafy adj.
枝叶茂密的

undergrowth n.
林下植被
into leafy undergrowth

midday n.
正午

clenches v.
紧咬
it clenches its jaws

jaws n.
双颚

vein n.
叶脉
a major vein underneath a leaf

dusk n.
黄昏
By dusk

lifeless adj.
无生命

rigidly adv.
牢牢地
the bite holds rigidly

muscles n.
肌肉

loosen v.
松开

fungal adj.
真菌

infectious adj.
具有传染性的
releases infectious spores

tread v.
行走
where foraging ants tread

词汇与表达（续）

invader n.
入侵者

fussy adj.
挑剔
is fussy about location

location n.
位置

bites n.
咬合点

north-west n.
西北

humid adj.
潮湿
the air is humid

abnormally adv.
反常
the fungus grew abnormally

disappeared v.
不见了

three-dimensional adj.
三维

microscope n.
显微镜

intertwined adj.
相互交织的
an intertwined lattice

lattice n.
网格
an intertwined lattice

evokes v.
使人想起
a picture that evokes

puppet n.
木偶

infiltrated v.
渗入
had not infiltrated the brain

narration n.
解说

inspire v.
启发了

makers n.
制作者

envisioned v.
设想

hypothetical adj.
假想的
The hypothetical scenario

scenario n.
情景
The hypothetical scenario

gamers n.
玩家

fictional adj.
虚构的

geological adj.
地质

fossil n.
化石
fossil leaf from Messel

characteristic adj.
典型的

scars n.
疤痕

文章简述

- 感染：泰国木匠蚁感染一种寄生真菌后离群，正午咬住离地约25厘米的叶背主脉，到黄昏死去。
- 传播：真菌破坏控制咬合的肌肉，使下颚无法松开，蚂蚁死后仍咬紧叶子；两三天后，蚁尸头部长出真菌细柄，向蚂蚁觅食的地面释放孢子，也就是可传播感染的微小颗粒。
- 环境：咬痕多在温湿度适宜的植物西北侧；把感染蚂蚁移高，真菌生长反常，把蚁尸放到地面，尸体则消失。
- 机制：2017年三维成像显示，真菌在腿和颚部肌肉周围织成网络，却没有侵入蚂蚁大脑。
- 边界：4800万年前的叶化石已有同类咬痕；记录这种现象的纪录片启发了游戏《最后生还者》，但游戏中真菌感染人类的情节是虚构的。